

Development of a Pipeline for Connectivity Analysis During EEG Microstates in Control Subjects and People with Multiple Sclerosis

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Abstract

This thesis project proposes an analysis of resting-state EEG microstates in healthy controls (HCs, $n = 24$) and people with multiple sclerosis (PwMS, $n = 51$), with the aim of identifying differences in functional patterns and their temporal dynamics. The pipeline begins with the extraction of individual maps using EEGLAB. Backfitting is then performed using the Grand Grand Mean as a common template. The analysis continues in FieldTrip for the study of functional connectivity and statistics. Orthogonalized amplitude envelope correlation (AEC) was used to estimate functional relationships between channels and reduce the effects of volume conduction. The resulting connectivity matrices provide the starting point for identifying connections that differ significantly between the two groups. The comparison is conducted channel by channel, applying the Mann–Whitney U test independently to each pair. The analysis continues with a graph-theoretical study in which nodes correspond to EEG channels and edges to connections. Centrality measures (degree, closeness, betweenness, and eigenvector centrality) characterize the role of each electrode in the network, while global measures (global efficiency, characteristic path length, and clustering coefficient) describe its overall organization. These measures are then summarized using descriptive statistics (mean and skewness) and subjected to statistical analysis using the Mann–Whitney U test and AUC calculated with the perfcurve function. The features that best distinguish the two groups are mainly related to betweenness centrality. Mean betweenness, which is higher in controls in almost all microstates (p between 0.023 and 0.048; effect size between 0.22 and 0.26; AUC between 0.67 and 0.65), is consistent with the reduction in betweenness centrality reported by Jouzizadeh et al. in people with multiple sclerosis and with Nauta et al., who link lower network integration in the delta band to a “linelike” configuration predictive of cognitive decline. Betweenness skewness, by contrast, is higher in patients (p between $5.665e-05$ and 0.03; effect size between -0.25 and -0.46; AUC between 0.66 and 0.79), in line with Carotenuto et al., who observed a redistribution of betweenness centrality across brain regions. The most significant result concerns microstate C and betweenness skewness ($p = 5.665e-05$). This microstate has been associated in the literature with the default mode network, consistent with Eijlers et al. In conclusion, analysis of microstates from EEG time series, combined with the study of functional interactions among channels, may provide a useful biomarker for understanding the development of multiple sclerosis.

Introduction

Multiple sclerosis is a chronic neurodegenerative disease of the central nervous system whose defining characteristic is demyelination: damage to the myelin sheath that surrounds axons and enables rapid transmission of neural signals. This alters both the speed of signal transmission and the overall organization of brain networks, which reorganize in response to the damage. Studying this reorganization requires tools that describe how different brain regions communicate. Resting-state EEG analyzed through microstates is well suited to this purpose. Microstates are brief topographic configurations of the brain's electrical field, stable for 30 to 120 milliseconds, during which the spatial distribution of the scalp signal remains almost constant before rapidly changing to another configuration. Studies combining EEG and functional magnetic resonance imaging have associated each microstate with the brain regions involved on the basis of the BOLD signal. In the literature, each microstate has also been associated with a specific resting-state brain network: microstate A with the auditory/phonological network, B with the visual network, C with the default mode network, D with the attention network, and E with the salience network. For the additional microstate in our study, G, these associations are less firmly established in the literature.

Functional connectivity differs from structural connectivity because it measures how two or more brain regions communicate over time, regardless of their physical connection. Studying connectivity separately for each microstate can therefore reveal which brain network is altered by the disease, information that might be lost if the EEG signal were analyzed as a whole.

Aim

The objective of this work is to study the functional connectivity associated with individual EEG microstates by comparing a control group (HCs, $n = 24$) with a group of people with multiple sclerosis (PwMS, $n = 51$). Specifically, the aim is to determine whether connectivity calculated separately for each microstate reveals differences between the groups both in network organization, through graph-theoretical measures, and in the temporal variables associated with each topography. More specifically, the work seeks to identify which combinations of frequency band, microstate, and graph measure show the most marked differences between the groups, to understand whether altered connectivity in multiple sclerosis affects the whole brain network uniformly or is concentrated in specific functional configurations.

Methods

The study involved 24 healthy controls (HCs) and 51 people with multiple sclerosis (PwMS). EEG signals were acquired using a 256-channel high-density cap distributed across the scalp. The most peripheral channels were excluded from the original 256, leaving 196 channels for analysis. These were subsequently divided into five regions of interest: frontal, central, parietal, occipital, and temporal.

EEGLAB and the MICROSTATELAB toolbox were used for microstate analysis. Functional connectivity was calculated with FieldTrip in MATLAB, while MATLAB and BrainNet Viewer were used for the graph-theoretical analysis.

The pipeline consists of three main stages: extracting microstate maps from the EEG signal, analyzing connectivity through amplitude envelope correlation (AEC), and applying graph-theoretical measures to the resulting matrices.

In detail, the topographic microstate maps were extracted using a modified k-means algorithm applied not to the entire continuous signal but only at Global Field Power (GFP) peaks. GFP is defined as the standard deviation of voltage across all channels at a given time. There are two reasons for this choice: the signal-to-noise ratio is better at GFP peaks, and applying k-means at every time point would be computationally excessive. Six centroids were chosen, in line with the reference literature, particularly Baldini et al. (2022). A trial with seven centroids had produced less promising results. To make microstate labels comparable across subjects and groups, the Grand Grand Mean shared template was calculated after the individual and group-mean maps, then sorted against the standard template of Koenig et al. (2002).

This procedure matches the resulting maps to those in the literature according to a similarity criterion, ensuring consistent labeling. Only after sorting was backfitting performed: each relevant time point in the original EEG signal corresponding to a GFP peak was assigned the label of the active microstate. For functional connectivity, each subject's EEG signal was first filtered into the frequency bands of interest: delta, theta, alpha, beta, and gamma. It was then divided into six derived signals, each obtained by aggregating the intervals during which a particular microstate was active.

Amplitude envelope correlation (AEC) was applied to these signals. For each pair of electrodes, the envelope was calculated using the Hilbert transform and, before Pearson correlation, each pair was orthogonalized in both directions to reduce the effect of volume conduction. Pearson correlation was then calculated between the orthogonalized envelopes of the two channels. The resulting value expresses how closely their amplitudes rise and fall together over time and represents functional connectivity between that pair of electrodes. Nearby electrodes can show high correlations simply because they partly detect the same signal, which propagates instantaneously through the head, rather than because of genuine neural communication.

The two groups were compared channel by channel using each subject's individual values. For each band and microstate, the subjects' connectivity matrices were grouped so that each connection had a set of values for each group. The Mann–Whitney U test was then applied separately to every connection, comparing the distribution of HC values with that of PwMS values for that channel pair. This yielded a p-value for each connection; the p-values were subsequently reassembled into a symmetric matrix.

Finally, for the graph-theoretical analysis, every EEG channel was represented as a node and every matrix connection as an edge. Graph measures were calculated from the connectivity matrix of each subject, band, and microstate. A weighted graph was built from each matrix, excluding self-connections. Four measures of nodal centrality were calculated: degree, closeness, betweenness, and eigenvector centrality. Degree measures how many strong connections a node has: the stronger its connections, the higher its value. Closeness and betweenness, in contrast, are based on distance between nodes rather than direct connection strength. The weight was therefore inverted so that a strong connection corresponds to a short distance between nodes. Eigenvector centrality

assigns a high value to nodes connected to other important nodes, rather than considering only the number or strength of their own connections. The clustering coefficient was calculated separately from the same connectivity matrix with its diagonal removed and normalized in the same way as the other measures.

These measures were selected to characterize both the role of each electrode in the network and the overall organization of connectivity. For each subject, each measure was summarized using two descriptive statistics, mean and skewness, to obtain one value per subject. These values were then compared between groups using the Mann–Whitney U test and MATLAB's `perfcurve` function, which calculates the ROC curve and the area under the curve (AUC), using PwMS as the positive class. An AUC above 0.5 therefore indicates that the feature better classifies patients, while an AUC below 0.5 indicates that it better classifies controls. All results were organized in a structure divided by microstate and populated with features: combinations of frequency band, descriptive statistic, and graph measure. The features for each microstate were ranked in ascending order of p-value to identify the combinations most relevant to distinguishing the two groups.

Results

The features that best distinguish the groups are mainly related to betweenness centrality, both its mean and its skewness. Mean betweenness is higher in controls in almost all microstates, with p-values between 0.023 and 0.048, effect sizes between 0.22 and 0.26, and AUC values between 0.67 and 0.65. Betweenness skewness shows the opposite pattern: it is higher in patients, with p-values between 5.665e-05 and 0.03, effect sizes between -0.25 and -0.46, and AUC values between 0.66 and 0.79. Taken together, these results indicate that betweenness is higher on average and more evenly distributed among nodes in controls, while communication in patients tends to depend on a smaller number of central hubs, with a more asymmetric distribution.

The overall pattern is consistent with independent studies in the literature. Jouzizadeh et al. found reduced betweenness centrality in people with multiple sclerosis. Nauta et al. linked reduced network integration in the delta band to a “linelike” network configuration predictive of cognitive decline, consistent with the more marked reduction in mean betweenness in the delta and theta bands in our study. Carotenuto et al. also observed a redistribution of betweenness centrality across brain regions in patients, in line with the increase in skewness found here.

The most significant result of the study concerns betweenness skewness in microstate C, with a p-value of 5.665e-05. Microstate C is associated in the literature with the default mode network. This result is consistent with Eijlers et al., who observed increased relative centrality of the default mode network in people with multiple sclerosis, suggesting reorganization of this network.

Regarding the temporal dynamics of microstates, comparison with Baldini et al. shows consistent trends, although they are not statistically significant. In our study, mean duration and occurrence increased for microstates A and B in patients and decreased for microstate D.

An important limitation concerns correction for multiple comparisons. Given the large number of tests performed in parallel, FDR correction was applied both to connections in the connectivity matrices and to graph-measure results. None of the findings just described survived this correction.

The reported p-values should therefore be considered uncorrected, and the identified features should be interpreted as exploratory results. Nevertheless, the trends are consistent with the cited studies, which observed similar patterns using different methods and tools.

Discussion of the Results

The results suggest a reorganization of functional networks in multiple sclerosis. In controls, communication between brain regions is more distributed, with betweenness spread more evenly among the different nodes of the network. In patients, communication tends instead to pass through a smaller number of central hubs, as indicated by increased skewness. This type of reorganization is compatible with the dynamics of multiple sclerosis: demyelination particularly affects long-range connections between brain regions, and a network that loses these connections tends to reorganize around a few central nodes rather than maintaining distributed communication across the whole network.

The most marked effect is observed in microstate C, which has been associated in the literature with default mode network activity. This suggests that the alteration does not affect the whole brain network equally but more strongly affects networks involved in higher-level cognitive processes, such as self-referential thought and introspection. This is also consistent with the non-significant trend in the temporal dynamics of microstates A, B, and D, suggesting that multiple sclerosis may change the behavior of individual microstates differently rather than acting uniformly.

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